



ChatGPT Unveiled: Understanding Perceptions of Academic Integrity in Higher Education - A Qualitative Approach

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Abstract

The purpose of this research is to gain a complete understanding of how students and faculty in higher education perceive the role of AI tools, their impact on academic integrity, and their potential benefits and threats in the educational milieu, while taking into account ways to help curb its disadvantages. Drawing upon a qualitative approach, this study conducted in-depth interviews with a diverse sample of faculty members and students in higher education, in universities across Lebanon. These interviews were analyzed and coded using NVivo software, allowing for the identification of recurring themes and the extraction of rich qualitative data. The findings of this study illuminated a spectrum of perceptions. While ChatGPT and AI tools are recognized for their potential in enhancing productivity, promoting interactive learning experiences, and providing tailored support, they also raise significant concerns regarding academic integrity. This research underscores the need for higher education institutions to carefully navigate the integration of AI tools like ChatGPT. It calls for the formulation of clear policies and guidelines for their ethical and responsible use, along with comprehensive support and training. This study contributes to the existing literature by presenting a comprehensive exploration of the perceptions of both students and faculty regarding AI tools in higher education, through a qualitative rich approach. By delving into the intricate dynamics of ChatGPT and academic integrity, this study offers fresh insights into the evolving educational landscape and the ongoing dialogue between technology and ethics.

Keywords ChatGPT · Higher Education · Academic Integrity · Faculty · Students

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Abbreviations

AI	Artificial Intelligence
GAI	Generative Artificial Intelligence
GPT	Generative Pre-trained Transformer
NLP	Natural Language Processing
LLMs	Large Language Models

Introduction

The globe has witnessed a fast-shifting environment in educational methods over the past decade. In fact, 98% of universities across the globe moved their classes online since 2020 (Higher Ed Partners, 2021). After the COVID-19 outbreak, higher education promptly shifted online, and distance learning has grown into standard and common practice (Kim & Kim, 2023). Even though the pandemic has hastened the transition to digital learning, this educational transformation began far before the pandemic as a result of technological developments (Xu & Ouyang, 2022). Among these developments, artificial intelligence (AI) has had the largest influence (Makridakis, 2017). Recent advances and expansions in machine learning have resulted in the creation of advanced and powerful digital resources capable of supporting education, such as generative artificial intelligence (GAI) (Mondal et al., 2023). One of the most controversial and innovative GAI models as Jovanovic and Campbell (2022) discussed is the Generative Pre-Trained Transformer (GPT). GPT models have drawn a lot of attention in the last couple of months especially following the introduction of OpenAI ChatGPT, an algorithm widely viewed as revolutionary (Mathew, 2023).

The ChatGPT technology processes and produces humanlike text using an extensive amount of accessible online data which is also known as natural language processing (NLP). Additionally, it can demonstrate originality in creating documents and dialogs effectively on various themes. In fact, ChatGPT is capable of engaging in human-like discussion and has been pondered to be adopted as customer care chatbot (Rivas & Zhao, 2023). Nonetheless, this technology has been firstly utilized in educational institutions by students worldwide (Javaid et al., 2023). The use of AI tools in education and scientific publishing, particularly ChatGPT, has sparked widespread debate and controversy across faculty, students, and researchers across the globe (Hill-Yardin et al., 2023). ChatGPT garnered mixed reactions from researchers and academics, echoing the history of debate over the advantages vs. hazards of emerging AI-technologies (Wogu et al., 2017). On the other hand, ChatGPT, like other GAI, can help with conversational and writing activities by increasing the efficacy and validity of the needed output (Deng & Lin, 2022). However, as Deng and Lin (2022) mentioned, concerns have been expressed, regarding probable bias due to the sets of data utilized in ChatGPT training, which might restrict its capacity and lead to incorrect information while ominously seeming scientifically reasonable. Furthermore, security considerations and the possibility of cyber-attacks with disinformation distributed through GAI has also been addressed (Sallam, 2023). Yet, as Trust et al. (2023) indicated, one of the main concerns that has sparked a furious outrage from faculty in higher education is academic integrity. As educational organizations and officials examine different approaches and strategies to address moral and ethical issues concerning the usage of ChatGPT and its counterparts, this study explores the perceptions and attitudes of students and faculty regard-

ing breaches of academic integrity upon utilizing ChatGPT. This research is of significant importance since minimal research has been dedicated to faculties and students' opinions regarding the usage of ChatGPT. Analyzing their perception is critical to ensuring student engagement and successful evaluation methods. A further gap that drove this study was the previous literature's inadequate examination of instructors' opinions of the obstacles and opportunities of applying ChatGPT, as well as the ways in which their understanding and practice of this method might be enhanced (Chen et al., 2020). To realistically urge instructors to practice and utilize ChatGPT, it is critical to grasp its drawbacks and obstacles from the perspective of insiders. Henceforth, this research employed a qualitative methodology to investigate the challenges and opportunities of applying ChatGPT in higher education from the perspective of faculty members and students.

Literature Review

Faculty Resistance

Previous studies have identified the fact that human intelligence is more dynamic than AI due to its physiological evolutionary background, versatility, inventiveness, emotional intelligence, and capacity to grasp complicated complex ideas (Korteling et al., 2021). Nevertheless, with the collaboration and coordination of both human intelligence and AI the results would be powerful and tremendously advantageous. Yet, the human mind's natural reluctance to undergo change and alter and adapt itself is a well-described phenomenon that may be understood from developmental and social psychology viewpoints (Tobore, 2019). Indeed, resistance to change and adaptability to these technological developments have been witnessed amongst faculty in higher education. Multiple reasons have been identified regarding faculty resistance including the unknown fears towards AI affecting academic integrity, including the rigid culture in academia, human's natural tendency to be uncooperative, as well as the lack of proper communication, information and trainings (Moerschell, 2009). Alongside, faculty may have limited exposure to AI tools and technologies, and their lack of familiarity can lead to apprehension (Ameen et al., 2002). In addition, faculty members are accustomed to having control over their teaching materials, methods, and assessments. Whereas, AI tools can sometimes appear as black boxes, making it challenging for instructors to understand how decisions are made or to customize content to their liking. This loss of control can be frustrating (Gulati et al., 2018).

Knowingly, the educational sector has witnessed technological advances and societal shifts that have led to digitalization throughout time. Although faculty and educational institutes remain hesitant to embrace AI's potential, yet the digital transformation's potential has an immense impact on engaging students while enhancing their overall involvement in higher education (Brunetti et al., 2020). Therefore, the ultimate resolution for universities to stay competitive while engaging students properly is through implementing a digital transformation strategy that addresses all faculty concerns and fears (Hashim et al., 2022).

ChatGPT in Educational Institutions

Researchers have discovered numerous advantages for the employment of ChatGPT in educational institutions, this includes individualized learning (Sok & Heng, 2023), stimulation of collaborative and engaging learning and teaching methods (Baidoo-Anu and Owusu Ansah, 2023), and the possibility of formative evaluation (Sallam, 2023). Nevertheless, ChatGPT has been highly criticized especially in the realm of science and academic institutions for causing a manifold of ramifications (Thorp, 2023). While ChatGPT is an exceptional program, this innovative technology has created substantial worries concerning how it would affect education. Claiming that while ChatGPT can produce articles on a variety of themes, its educational content is still in its early stages (Biswas, 2023). As a result, cheating has been one of the results of the utilization ChatGPT forcing academics to reevaluate their courses using novel techniques and issue tests that are difficult for AI to answer (Lund & Wang, 2023). Consequently, this has led to the immense attraction of contract cheating (commercial ghost writing) in media outlets and higher education publications. At the core of the problem is the risk of image harm to institutions, professions, and the industry, as well as public safety, if students lack the necessary skills and capacities (Stone, 2022). On the contrary, certain instances where faculty or students utilize these tools include assisting students in generating persuasive marketing copy and crafting compelling product descriptions, providing instant feedback on language use and style for marketing students. Or even, in international business studies, ChatGPT's language translation capabilities aid students in translating business documents, contracts, and marketing materials into different languages for global business scenarios. Whereas, a professor might employ AI algorithms to recommend specific case studies or supplementary materials to students based on their progress and areas of interest.

ChatGPT and Plagiarism

Academic plagiarism has long been prevalent in educational institutions across the world (Berlinck, 2011). Although cheating isn't an entirely novel notion, some students referred to outsourcing educational assessment for money via a third party which has grown increasingly common or possibly improved as technological innovation has advanced (Awdry & Ives, 2022). As a result of the interruption to on-campus academics caused by COVID-19, there was a swift shift towards online learning and evaluation systems, which broadened the possibilities for cheating (Hill et al., 2021). Additionally, students' ability to access a wide sea of knowledge due to the internet has made it extremely enticing for some to indulge in unethical methods such as copy-pasting or paraphrasing without proper reference (Comas-Forgas & Sureda-Negre, 2010). Traditional plagiarism detection tools, such as Turnitin and other anti-plagiarism software, have proved beneficial, but they frequently fail to detect increasingly sophisticated kinds of plagiarism, such as AI-generated material (Xiao et al., 2022). In fact, ChatGPT is capable of producing text that closely mimics human writing, increasing educators' difficulties in combating academic dishonesty (Fitria, 2023). One of the most significant fears concerning ChatGPT's impact on academic plagiarism is the possibility of student exploitation. Some students may resort to utilizing ChatGPT to write essays, papers, or other academic projects rather than investing time and effort in performing legitimate research and creating unique material. This raises ethical concerns

concerning academic work's authenticity and integrity. As the line between student effort and AI-generated work blurs, educators and institutions face the problem of efficiently identifying and resolving plagiarism. In fact, the abuse of ChatGPT has resulted in an unequal academic playing field. In a way that, students who utilize ChatGPT to create original content may acquire a disproportionate benefit over other students who have limited access to it or decide against employing it for ethical reasons (Cotton et al., 2023). This difference has the potential to distort grades and academic praise, surpassing the merit of diligent work and dedication to oneself.

Methodology

Due to the limited literature on the perspective of educators and students on AI tools and their effect on academic integrity, the present study adopted a qualitative research method. This method was utilized due to its ability to offer the study an in-depth data collection from individuals (Donald et al., 2010). The study was conducted using one-on-one interviews at different universities in Lebanon. This study included students and faculty from several universities with different levels and expertise. Participants were approached utilizing a convenience sampling technique, the students and faculty interviewed were based on the convenience, accessibility and willingness to participate. In order to ensure representation from an array of experience and expertise, participants were chosen based on their level of familiarity with AI tools. In fact, a variety of issues were evaluated: the participants' familiarity with the recently introduced ChatGPT, its applications, facilitators and the impediments to its implementation into their academic experience. Additionally, efforts were made to include participants from a variety of educational backgrounds and contexts in order to enhance the data and enable a thorough analysis. The technique utilized was efficient, less costly, and less time consuming compared to other techniques (Suen et al., 2014). The sample included twenty faculty members, and twenty students each having a different experience with ChatGPT. As a matter of fact, based on the data set presented, saturation of data has been achieved within these 40 interviews, taking into account that the basic elements for meta themes were present as early as twelve interviews, where further interviews did not yield additional insights.

The interview protocol consisted of four structured questions addressed to both students and faculty members. Covering their comprehension of AI tools, methods for upholding academic integrity, and views of AI technologies as either facilitators or threats to academic integrity. The same questions were posed to all the participants in precisely the same order to ensure uniformity and facilitate a credible comparative analysis.

Students' Questions

- Are you aware of the new AI module ChatGPT? Have you ever used it before?
- How do you ensure that you maintain academic integrity and avoid plagiarism when incorporating information generated by ChatGPT into your assignments or research papers?
- Do you view ChatGPT and similar AI tools as a threat to student academic integrity?

- Do you believe that ChatGPT and similar AI tools have the potential to become a helpful tool instead of a threat?

Faculty Questions

- Are you aware of the new AI module ChatGPT? Have you ever used it before?
- What is your understanding of plagiarism, and how do you think the use of ChatGPT or similar AI tools could potentially lead to unintentional plagiarism in academic work?
- Do you view ChatGPT and similar AI tools as a threat to student academic integrity?
- Do you believe that ChatGPT and similar AI tools have the potential to become a helpful tool instead of a threat?

The initial screening and piloting of the study was conducted in August 2023, whereas the data was collected during September, 2023. The Institutional Review Board of the attending university granted ethical approval (LAU.SOB.SK8.4/IRB). Prior to their enrollment in the research and recording, all participants provided a written permission. The goal of the study, the voluntary nature of their involvement, and their freedom to withdraw at any moment were all explained to participants. Pseudonyms were employed to preserve the participants' anonymity. The questions were sent via email and answered via email or booked a one-on-one interview session. One-on-one interviews lasted on an average between 8 and 10 min. Which were tape recorded, and verbatim responses to each question were transcribed utilizing standardized transcription protocol to ensure precision (McLellan et al., 2003). Additionally, the entire interview discussions were conducted in the English language to maintain consistency across interviews.

While all interviews were transcribed, anonymized, and allocated the participant IDs prior to analysis. A thematic analysis of the interview transcripts was conducted (Braun & Clarke, 2006). The first step in the analysis was getting acquainted with the interview transcripts. Going over each transcript more than once to make sure we understood the context and substance completely. To aid in the categorization and analysis that followed, the initial impressions and notes were documented. After familiarization, preliminary codes were created to identify the main ideas, concepts, and recurrent themes found in the data. The application of codes was done both inductively, which allowed for the creation of new themes from the data, and deductively, which was based on pre-established themes identified from the interview process. We utilized an inductive qualitative method to create a collection of descriptive theme memos that explained each category using interview data as a foundation (Saldaña, 2021). The initial codes were categorized into overarching themes and sub-themes by means of an ongoing process of comparison and assessment. Throughout this iterative process, the connections between codes were closely examined, participant and interview response similarities and differences were investigated, and the thematic framework was adjusted as necessary. The memoranda were then debated and amended on an ongoing basis by all authors, revealing the links amongst the categories and eventually clarifying the themes, resulting in the five major findings presented below. The coding processes included manual coding methods and data management and organizing using NVivo software. The capabilities of NVivo, including matrix coding and query tools, were utilized to investigate

connections between codes and themes, spot trends, and provide visual representations of the data. To make sure the themes appropriately represented the viewpoints and experiences of the participants, they were examined in light of the original interview data. The resulting theme framework served as the foundation for further interpretation and analysis as it offered a thorough and nuanced grasp of the research topic.

Findings

In order to acquire a thorough extensive understanding of how students and faculty view ChatGPT and their perspective on academic integrity, a comprehensive set of questions were designed into interviews each subjected to faculty and students separately. After analyzing the data coded, 5 themes were concluded and coded accordingly; Awareness and Usage, Maintaining Academic Integrity, Threats of ChatGPT, Benefits of ChatGPT and AI Tools, AI Tools' Potential as Aids.

Primarily, in order to grasp an idea of the most frequent words mentioned in the interview by interviewees we created a word frequency cloud via NVivo, that gave us an idea about recurring words that has helped us while creating our codes and themes. (Fig. 1).

Awareness and Usage

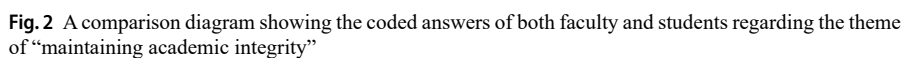
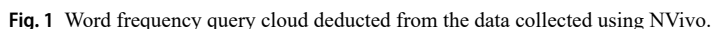
Unlike other questions, all students and faculty had similar answers that point out the definite awareness of ChatGPT. Nevertheless, only one student and one faculty member replied that they are aware of ChatGPT yet they have never used it unlike the rest of the interviewee's where they use this tool.

"I am aware but never used it" (Student 7).

"Yes, I am aware but didn't use it." (Faculty 9).

Maintaining Academic Integrity

In point of fact, it has been clearly identified that academic integrity and plagiarism reveals a nuanced approach among students in utilizing AI tools like ChatGPT. It is evident that both students and faculty are acutely aware of the potential threats to academic integrity posed by these tools, with plagiarism emerging as a main concern (see Fig. 2). On the other hand, students demonstrate a proactive approach to ethical usage. They acknowledge the risk of using ChatGPT for academic purposes and actively employ strategies to mitigate it, primarily through paraphrasing the AI-generated content to ensure it aligns with their own work. The additional insight provided by some students, ensure that they paraphrase to further disguise the AI-generated information, underscoring their commitment to avoiding ethical issues while taking advantage of the benefits of AI. This multifaceted approach not only showcases students' responsibility but also highlights the evolving landscape of academic integrity in the digital age, where students are taking an active role in preserving the authenticity of their work while engaging with AI tools. Overall, we understood that students actually utilize ChatGPT information yet they paraphrase the data received from AI tools to avoid plagiarism.



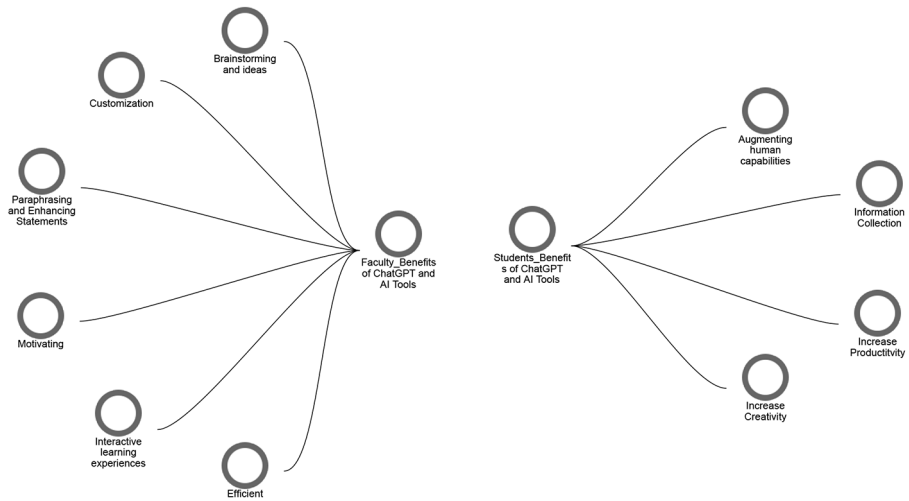


Fig. 3 Comparison chart between faculty and students regarding the benefits of ChatGPT and other AI tools, via NVivo.

“If I ever need to use ChatGPT for academic purposes, I make sure I understand the gist of whatever info the AI tool generates for me and word it in my own way (basically paraphrase as needed).” (Student 6).

I would make sure that I paraphrase and reword everything while ensuring correct information used.” (Student 13).

Nevertheless, an underestimated response was given by 2 students stating that they use additional AI tools that help them paraphrase, stating that they paraphrase in order to avoid unethically while using other AI paraphrasing tools.

“I use other tools for paraphrasing, so in this case I ensure that the information will not be detected as AI generated info.” (Student 3).

Similarly, the faculty’s responses collectively highlight their commitment to upholding academic integrity in the face of AI integration in education. Faculty emphasize the necessity of paraphrasing and proper citation as essential ways against plagiarism when utilizing AI tools like ChatGPT. They define plagiarism as the direct utilization of someone else’s work without due attribution and stress the importance of referring to credible sources to maintain academic integrity. Notably, it came to light that faculty members appear to be unaware of AI tools designed to aid in paraphrasing, potentially leaving students with minimal or no personal effort. Nevertheless, the faculty’s emphasis on synthesizing findings and utilizing diverse sources to strengthen arguments highlights their dedication to preserving academic integrity and ensuring that students produce original, well-researched work, even as technology reshapes educational practices. This collective faculty perspective stresses on the

need for institutions to provide comprehensive guidance and training to both faculty and students on the responsible and ethical use of AI tools in academia.

“Not only chat GPT should be used in research, other sources as well as classic research tools and techniques are still helpful. Information has to be revised, rewritten, paraphrased and customized to fit researcher’s objectives using one’s own words to avoid plagiarism.” (Faculty 2).

“Copy and pasting as is from chatgpt would be considered plagiarism as this will not be your work but the work of an algorithm ... You are not supposed to copy and paste but you need to paraphrase and cite your source.” (Faculty 3).

“They still have to synthesize their findings and find eligible resources to make their arguments strong.” (Faculty 1).

“Students can’t be restricted from using developed AI tools, yet they should follow proper guidelines on how to use these tools to avoid plagiarism...” (Faculty 18).

(McLellan, MacQueen, and Niedig 2003) After the interviews has been conducted, the themes had been saturated. Thematic analysis was performed to assess the data, which included predefined topics as well as enabling new themes to arise from the data. The transcripts were reviewed several times to detect trends and commonalities in the data. Based on the study objectives, a coding system was developed and then adapted to the data utilizing NVivo 12 software(15). The research team selected and clarified themes via a series of iterations of coding, examining, and debating the data until consensus was achieved. The Institutional Review Board of the attending university granted ethical approval (Ref. No. 23/0155/IRB). Prior to their enrollment in the research and recording, all participants provided written permission. The goal of the study, the voluntary nature of their involvement, and their freedom to withdraw at any moment were all explained to participants. Pseudonyms were employed to preserve the participants’ anonymity.

As previously mentioned, Fig. 2 reflects multiple common matters that both students and faculty take into account in order to maintain academic integrity. This includes paraphrasing upon retrieving information from AI tools. In addition to utilizing AI detection tools that can help detect whether the information stated has been composed directly from AI tools without paraphrasing. Moreover, the utilization of credible sources and scholar citation are highly recommended by faculty. Whereas, it has been indicated that students are eager to work on assignments and exams that require critical thinking that AI tools can’t help with.

Threats of ChatGPT

The third question investigated how students and faculty perceive ChatGPT. The answers provided by the students collectively shed light on the multifaceted nature of AI tools in the educational context. In reality, the dichotomy of these tools was acknowledged, emphasizing that their impact can vary significantly depending on how they are used. Whereas, we were able to interpret that when wielded irresponsibly, AI tools may become a source of concern. This includes the potential for these tools to encourage plagiarism, hinder the development of critical thinking skills, and even propagate biases, emphasizing the importance of ethical and thoughtful utilization. However, Student 1 also underscores their potential to enrich the learning experience and facilitate research when approached with responsibility and

discernment. On the other hand, student 6 delves into a different aspect of the issue by drawing attention to the prevailing attitudes among students regarding academic integrity. The quote underscores a troubling trend where a significant portion of students may prioritize convenience over ethical academic practices. The perception that AI tools like ChatGPT are undetectable adds to the concern, suggesting that such tools might inadvertently encourage unethical behavior by providing an apparent shortcut to academic success. Another alarming dimension to the discussion was brought to light by highlighting the notion that AI tools pose a threat when students over-rely on them. This perspective underscores the importance of maintaining a balance between leveraging AI tools as educational aids and ensuring that students develop essential critical thinking and problem-solving skills. It suggests that excessive dependence on AI tools may hinder the educational growth of students. Lastly, a more practical concern was extracted regarding the reliability of AI-generated information, emphasizing the necessity of caution when relying on AI tools for information gathering, research, or academic work. Subsequently, this draws attention to the need for students and educators to critically evaluate and cross-reference information obtained from AI tools to ensure its accuracy and credibility.

In sum, these quotations collectively emphasize the intricate interplay between the benefits and potential pitfalls of AI tools in education. They underscore the pivotal role of responsible usage, ethical considerations, and the development of critical thinking skills in harnessing the full potential of AI tools while safeguarding academic integrity.

"If used incorrectly they may lead to plagiarism, hinder critical thinking, and propagate biases. However, when used responsibly, they can enhance learning and research." (Student 1).

"...Not many students take academic integrity seriously and a lot of them would rather take the easy road especially if they know that the AI tool they use is undetectable." (Student 6).

"When students depend on it without putting any extra effort them it's definitely a huge threat" (Student 16).

"I think that ChatGPT and similar AI tools have the potential to be a threat to student academic integrity if they are not used responsibly." (Student 12).

"... I even tried to gather credible information from it, yet it gave me fabricated information on fake research." (Student 8).

On the other hand, faculty had all similar answers pointing out to the way ChatGPT is utilized is what positions AI tools as a threat or as a beneficial tool. Nevertheless, all faculty pointed out to the importance of initiating university guidelines and conducts that include proper way of utilizing ChatGPT and similar AI tools. Having a clear code of conduct for such tools, and giving clear instructions of the proper utilization for both students and faculty. In turn, this will ensure ethical use, promoting responsible behavior among faculty, staff, and students.

When examining the viewpoints of faculty members regarding AI tools like ChatGPT in education, a common theme emerges: a belief in the compatibility of these tools with academic integrity when used appropriately. In fact, ChatGPT is considered as a tool that doesn't inherently threaten student integrity, emphasizing the importance of proper citation and acknowledgment of AI-generated content. Similarly, it has been identified that there is

a significance when it comes to the adherence to university guidelines and policies for using AI GPT tools, asserting that academic integrity can be preserved within these parameters.

Collectively, these faculty perspectives reflect a sense of trust in students' ability to incorporate AI tools as valuable aids in their academic endeavors while upholding the principles of academic integrity, provided they follow established guidelines and practices.

"No, I don't view it as a threat to student integrity, as long as the student cites his/her work and mentions what was taken from ChatGPT. It is just another tool, with easier features." (Faculty 4).

"Academic integrity is not violated as long as the usage of such tools abide by the university guidelines/policy for using AI GPT tools." (Faculty 7).

Benefits of ChatGPT and AI Tools

A compelling array of advantages emerged upon mentioning the benefits of ChatGPT and other AI tools (Fig. 3). In fact, students highlighted its role in augmenting human capabilities, facilitating information collection, boosting productivity, and nurturing creativity, showcasing its value as an educational and productivity tool. On the other hand, faculty members emphasized ChatGPT's prowess in paraphrasing and enhancing statements, igniting brainstorming sessions, tailoring feedback to individual needs, fostering motivation, and creating interactive learning experiences, all contributing to an efficient and dynamic educational environment. Together, these perspectives underscore the versatile and transformative potential of ChatGPT in both academic and professional settings, as it empowers users to harness AI-driven capabilities for enhanced learning, creativity, and productivity.

AI Tools' Potential as Aids

The final interview question inquired about the opinions of both students and faculty members regarding their perspective of ChatGPT as a helpful tool. Summing up both faculty and students, even though some complained about it as a threat, all agree that it is a beneficial and a crucial tool for the future. Since both students and faculty view ChatGPT as a valuable tool, it becomes apparent that there is a consensus regarding its beneficial aspects and its potential importance for the future of education. The consensus centers on the advantages of increased productivity, support for learning, enhanced creativity, and assistance in research. This indicates a general recognition of the positive impact of AI tools like ChatGPT in educational settings. However, there is a consistent emphasis across all interviews on the importance of responsible and ethical usage, underscoring the need for a balanced approach to incorporating these tools into educational practices.

"Yes, they can increase productivity, support learning, increase creativity, and help in research." (Student 1).

"It is awesome saving time and gathering information... It even allows me to invest time somewhere else" (Student 8).

"... ChatGPT has helped me tremendously while doing my research".

“Absolutely yes, I can see how these tools are being widely used by industry to speed up work processes... These tools are similar to an assistant that produce a first draft of any work that needs to be enhanced professionally.

The refusal of such tools brings back to my mind the demonstrations against students’ usage of calculators in the 1920s...” (Faculty 7).

“Well definitely these tools could be very helpful we cannot escape them. This is the future, like any other instrument if it’s used in the correct manner ...” (Faculty 11).

“It is a great advantage to help in productivity and efficiency. Allowing faculty to focus on more creative and innovative subjects while doing their tasks.” (Faculty 6).

Discussion

The main objective of this study was to examine students’ and instructors’ opinions of academic integrity in the setting of AI. This research used interviews to collect responses to the primary research questions. According to the findings of the interview, all faculty members believe in the beneficial nature of AI technology. Yet almost all faculty insist on the manner of usage as a judgment to whether it is a threat or a beneficial tool. In addition, students actually approve of faculty’s judgment on ChatGPT, shedding the light on paraphrasing as a way to not indulge in plagiarism and fall into the trap of breaching academic integrity. Moreover, this research recognized that educational agencies and institutions must remain up to date on AI advances and modify their applicable regulations and policies correspondingly. These findings are consistent with prior research by Lim et al. (2023) and Mohamed (2023). Reiterating that employing AI-generated information for writing assignments and completing tasks explicitly qualifies as academic misconduct (i.e., plagiarism) and revising applicable regulations in accordance must be accompanied by examples that demonstrate how students can utilize AI tools to generate inspiration, establish preliminary concepts, and review primary sources of information. In fact, this has been clearly stated by one of the students:

“One way that schools and universities should use AI is by emphasizing the importance of analysis and critical thinking in students. By having such skills, students can acquire knowledge that maybe some AI tools could not obtain. Another way that could be used is to use such AI tools in teaching students to properly use it by gathering information and ideas instead of solving and grading their assignments or papers.” (Student 10).

The second section of the interview examined the influence of AI tools on academic integrity and was consistent with the findings of Perkins (2023). Indicating that LLMs provide an obvious potential danger to academic integrity since instructors may be unable to accurately determine the quantity of information generated by a student and hence offer an appropriate evaluation of a student’s grasp and interpretation of the issue at hand. This is the primary issue that was identified when asking students about plagiarism, where they were abrupt to point out that they do withdraw content and information from ChatGPT, yet they paraphrase it. However, it has been identified that students even utilize other AI tools to help them with

paraphrasing. This serves as a future need for investigation as it reflects a further ethical concern.

The third section of the interview looked into the impact of employing artificial intelligence and its considerations as a threat leading to academic dishonesty. Without a doubt, faculty and students uniformly agreed that AI had a negative impact on students' academic integrity if used unethically (plagiarism). They believed that artificial intelligence has boosted the accessibility and attraction of academic dishonesty for students, limiting the development of essential universal and transferrable abilities. In relation to the aforementioned, they emphasized the vitality of proper citation and paraphrasing. As a result, most instructors raised the claim that it is critical for institutions and politicians to address the ethical consequences of AI-driven academic dishonesty as soon as possible. Furthermore, the interviews revealed that professors' failure to detect AI-generated assignments is another factor leading to students participating in academic dishonesty. These findings are consistent with those of other research (Fitria, 2023; Dergaa et al., 2023; AlDhaen, 2022), in which there was widespread worry about the incidence of academic dishonesty. Nonetheless, this viewpoint differs from the one that labels AI as a "double-edged sword" (Shah, 2023), because, despite certain obstacles that accompany AI implementation, participants of this study, along with participants in various other studies, vastly acknowledged the various advantages that AI provides to students.

Finally, the last section of the interview addressed whether ChatGPT could be considered as a beneficial tool. As previously mentioned, instructors and students have acknowledged its benefits with proper guidance and ethicality in the sense that students must properly utilize these tools through citation and paraphrasing. Alongside, it has been mentioned that AI advancements are undeniable and either way they are embedded in all aspects of society. As a faculty member related; *"The refusal of such tools brings back to my mind the demonstrations against students' usage of calculators in the 1920s."* (Faculty 7). Hence, both instructors and students emphasize on the importance and expediency of AI tools such as ChatGPT. Nonetheless, its practical utilization is what claims it as a double-edged-sword.

Implications

Ultimately, the advent of ChatGPT and other Language Model AI technologies promise a rich tapestry of advantages for higher education, heralding greater student engagement, collaboration, and accessibility. ChatGPT's utility in asynchronous communication, timely feedback provision, facilitation of student group dynamics, and support for remote learning is invaluable. Additionally, its capabilities extend to language translation, summarization, question answering, text generation, and customized assessments, revolutionizing the educational landscape. These innovative tools also pave the way for adaptive learning, tailoring educational experiences to individual students, and AI-powered mentorship, providing 24/7 student support. They enhance accessibility by making educational materials more inclusive and can fuel data-driven decision-making to continuously improve courses and programs. Furthermore, AI can serve as a catalyst for ethical AI education, ensuring that students and faculty are not only proficient in AI technologies but also aware of the ethical implications and responsibilities associated with their use. With global collaboration opportunities, AI can foster international connections among universities and students from diverse backgrounds. AI-driven creative expression and AI-assisted research collaboration can also

expand horizons. However, amid this potential lies a complex web of challenges, most notably the thorny issues of academic integrity and plagiarism. As ChatGPT and AI tools become increasingly integrated into educational settings, they create fertile ground for academic dishonesty, blurring the lines between human and machine-generated content. In this intricate landscape, universities stand at a crossroads, tasked with a weighty responsibility. They must meticulously weigh the risks and rewards of deploying these technologies and, in doing so, chart a course towards ethical and responsible use. This journey necessitates the formulation of clear rules and protocols for implementation, the provision of comprehensive training and support for both students and faculty members, and the adoption of a multi-faceted approach to detect and prevent academic misconduct. By addressing these complex issues head-on, universities can harness the full potential of AI tools while steadfastly upholding the integrity of their academic assessments and the quality of their educational offerings. In doing so, they not only navigate the ever-evolving landscape of technology in education, but also serve as beacons of responsible innovation, ensuring that the core values of academic excellence and integrity endure in the digital age. Furthermore, it has come to our notice that every time we discover a disadvantage in AI tools, promptly another AI model is created to solve the disadvantage of the initial problem found. This brings to light the idea that while the human nature resists change due to ethicality issues, AI engineers on the other hand are being extremely productive through finding solutions. For instance, a student highlighted models that help in paraphrasing content without any effort. This, in turn, makes us question the need for paraphrasing information extracted from AI language models, when there is another AI tool to do so. The right structure to approach such tools is through harnessing their benefits and enhancing human creativity without affecting the remarkable function of the human brain.

Limitations and Future Research

Despite the comprehensive nature of this study, several limitations should be acknowledged. Primarily, the sample size is limited to Lebanon. Which offers a unique perspective, as Lebanon is experiencing a crisis marked by a severe economic recession and political instability. Nevertheless, this restricts the findings' generalizability to other regions. Studies that compare several rising economies with comparable socioeconomic issues may yield important information about the uptake and effects of AI technologies in various settings. By exploring the experiences, challenges, and successes of these regions, researchers can uncover valuable insights into how AI is reshaping education in resource-constrained environments and how it intersects with academic integrity. Additionally, it can shed light on the strategies and measures that institutions and policymakers in emerging economies employ to ensure academic integrity in the era of AI-driven education. Secondly, convenience sampling presents possible biases given that it might not fully represent the larger population, despite being a practical method. Future research could use random sampling techniques to reduce this bias and offer a more representative sample. Thirdly, qualitative methods cannot investigate causal relationships between variables in the same way quantitative methods can. In a way that, qualitative research does not exclude alternative explanations for conclusions regarding causality. Hence, it would be beneficial for future studies to use mixed-methods approaches, combining qualitative and quantitative data to give a more comprehensive picture of how AI technologies such as ChatGPT affect academic integrity.

Finally, even though this study sheds light on important topics like usage, awareness, and academic integrity in relation to ChatGPT, more research on the moral ramifications and the creation of frameworks and norms for the appropriate application of AI in education is needed. Gaining insight into the viewpoints of other stakeholders, such as tech developers and legislators, may also help to better comprehend the benefits and difficulties that artificial intelligence in academia presents. Future research can build on the results of this study by addressing these constraints in order to gain a deeper understanding of the intricate relationship between academic integrity and AI tools. This understanding will ultimately help to guide the responsible and successful integration of AI in higher education.

Conclusion

In conclusion, this research journey has taken us through a profound exploration of ChatGPT, a remarkable chatbot that has captured significant attention in recent times. It has traversed the intricacies of ChatGPT in the context of higher education, shedding light on faculty resistance and delving deep into its historical evolution, technological underpinnings, and transformative potential. Through an exhaustive analysis, we've unveiled five pivotal themes that highlight the dynamics of ChatGPT in academia, ranging from awareness and usage to the critical nuances of maintaining academic integrity. The emergence of AI, including the likes of ChatGPT, within the academic realm has ignited a crucial discourse on academic integrity. As AI-driven tools permeate the educational landscape, they force us to reevaluate the very essence of knowledge creation and assessment. This sparks essential questions about the ethics of technology's role in academia, raising concerns about the thin line between assistance and plagiarism. As we move forward, it becomes imperative for higher education institutions to navigate this evolving landscape with nuance and foresight. We must explore innovative ways to harness the benefits of ChatGPT and similar AI tools while mitigating the potential pitfalls, all while maintaining the core principles of academic integrity. Ultimately, this research invites us to ponder not only the impact of AI on academia but also the profound questions it raises about the nature of knowledge, originality, and the future of education in an increasingly AI-enhanced world.

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Data Availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing Interests The authors declare that they have no competing interests.

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